

UNARY RELATIONAL OPERATION.

The rename operator is denoted by $\rho(\rho)$.
In some case we might want to rename.

- (i) attributes
- (ii) relation name } or BOTH.

The general rename operation is expressed as.

$$\rho_S(B_1, B_2 \dots B_N)(R) \text{ --- (a)}$$

this changes the relation name R to S .
the column attributes to $(B_1, B_2 \dots B_N)$

$$\rho_S(R) \Rightarrow \text{only changes the relation name} \text{ --- (b)}$$

$$\rho(B_1, B_2 \dots B_N)(R) \Rightarrow \text{changes only the column names} \text{ --- (c)}$$

Relational Algebra Operation from Set theory : UNION.

Binary operation, denoted by $\cup$.

The result of $R \cup S$ is a relation that includes

- (i) all the tuples that either in R or in S or in both $R \& S$.

* (Note that the two relations should be union compatible).

- (ii) Duplicate tuples are eliminated.

* Union compatible means that each pair of corresponding attributes must be type compatible.

EXAMPLE:

Retrieve the SSN of all employees who either work in department 5 (Result 1 below) or directly supervise an employee who works in department no. 5 (Result 2 below).

DEPS_EMP $\leftarrow \sigma_{DNO=5} (EMPLOYEE)$

RESULT 1 $\leftarrow \pi_{SSN} (EMPLOYEE)$

RESULT 2 $\leftarrow \pi_{SUPERSSN} (DEPS_EMPS) *$

RESULT $RESULT 1 \cup RESULT 2$

TYPE COMPATIBILITY:- an important property required to execute set operations.

$R_1(A_1, A_2, \dots, A_n), R_2(B_1, B_2, \dots, B_n)$

if ;

- (i) they have the same no. of attributes
- (ii) the domain of each corresponding attribute is also the same.

$$\text{dom}(A_i) = \text{dom}(B_i)$$

PROPERTIES OF SET THEORY R.A:-

- (i) They are (union & intersection) commutative.
 $R \cup S = S \cup R, S \cap R = R \cap S$

- (ii) They can be considered as n-ary operations as they can apply to any no. of relations due to associative nature

$$R \cup (S \cap T) = (R \cup S) \cap T$$

(iii) The minus operation is not commutative.

→ cross product.

SET THEORY RA (CARTESIAN PRODUCT).

Combines tuples from two relations in a combinational fashion.

Such that.

$$R(A_1, A_2 \dots A_n) \times S(B_1, B_2 \dots B_m)$$

$$Q = (A_1 A_2 \dots A_n, B_1 B_2 \dots B_m).$$

for each combination

✓ The resulting relation has 1 tuple from R and 1 from S. Degree of the relation is $n+m$ attributes.

Note; if R has $|NR|$ tuples & S has $|NS|$ tuples; the $R \times S$ will have $n_r * n_s$.

Note usually the cartesian product may not be meaningful unless applied with a filter.

Not meaningful example

$$FEMALE_EMP \leftarrow \sigma_{SEX='F'}(EMPLOYEE)$$

$$EMPNAME \leftarrow \pi_{FNAME, LNAME, SALARY}(EMPLOYEE)$$

$$EMP_DEPENDANT \leftarrow EMPNAME \times DEPENDANTS$$

Meaning full example.

//

//

$$EMP_DEPENDANTS \leftarrow EMPNAME \times DEPENDANTS$$

$$ACTUAL_DEPS \leftarrow \sigma_{SSN=ESSN}(EMP_DEPENDANTS)$$



BINARY RELATIONAL OPERATION JOIN.

This operation is very important for the relational databases with more than a single operation, as it allows us to combine related tuples from various relations.

General form

$R \bowtie_{\text{join condition}} S$

→ Does the join of Cartesian + select in single move!

Example: Retrieve the name of the manager of each department.

$\varphi \leftarrow R \bowtie_{SSN = MGRSSN} S$

BINARY OPERATION (NATURAL JOIN).

Natural join is used to automatically join the tables based on columns with the same name.

While not duplicating columns like EQUJOIN.

example 1.

(a)

(b)

$DEPT_LOC \leftarrow DEPARTMENT \star DEPT_LOCATIONS$

consider both (a) & (b) has a common col Numbers so the join through $\star$ will automatically:

Department. Number = DEPT-LOCATION. NUMBER.

example no. 2

$\sim x \sim$

$\varphi \leftarrow R(A, B, C, D) \star S(C, D, E)$

$\varphi \Rightarrow R.C = S.C \text{ AND } R.D = S.D$

Note if no, cols have the same name we first have to rename our column.

DIVISION RELATIONAL OPERATION

It answers questions like:

find all the employees who work on projects that smith works on.

Working: $C \leftarrow R \div S$

- (i) group the dividend relation (R) by X attributes
 - (ii) for each X attribute, collect all its associated y values.
 - (iii) keep only those X whose $\uparrow$ set contains all y values from the divisor.
- (similar to projection + diff etc.)

Aggregate Functions & Grouping.

mathematical aggregate function that apply on a collection of values of database.

f MAX salary (EMPLOYEE)

f COUNT SSN, AVERAGE SALARY (EMPLOYEE)

EXAMPLE (PROCEDURAL FORM).

Qno. 1: Retrieve the name and address of all employees who work for the research department.

Research_Dept $\leftarrow \sigma_{DName = 'Research'}$ (EMPLOYEES)

* Research_EMPID $\leftarrow$ Research_Dept $\bowtie$ EMPLOYEE
DNO. = DNUMBER

Qno. 2: Retrieve the names of employees who have no dependants.

$ALL_EMP \leftarrow \pi_{SSN} (EMPLOYEE)$
 $EMPS_WITHDEPS (SSN) \leftarrow \pi_{ESSN} (DEPENDANT).$

$EMPS_WITHOUTDEPS \leftarrow (ALL_EMP - DEPENDANT)$
 $RESULT \pi_{LNAME, SNAME} (EMP_WITHOUT_DEPS).$

CHAPTER NO. 14:-

Q. Explain Redundant Information in Tuples and (Update) anomalies.

Consider the relation EMP_Proj (Emp#, Proj#,ENAME, Pname, No_hrs).

- (i) Update anomaly: Changing name of project number P1 from "Billing" to "Non-Profit" may cause this update to be made to all 100 employees ^{working} on the project.
- (ii) Insert anomaly: Cannot insert a project unless an employee is assigned to.
- (iii) Inversely cannot insert an employee unless an he/she is assigned to a project.
- (iv) Delete anomaly: Deleting a project may delete all the employee working on that project and vice versa.

Q. Explain the concept of spurious Tuples.

Spurious are misleading tuples that appear when you join poorly designed tables.

⇒ loss less join is thus a property that when we split our data across (Normalization) and then join them back together, we get exactly the original data with no loss.

Q. Describe the goals of Normalization:

- (i) Semantics of the Relation Attributes.
- (ii) Reducing the redundant values in tuples.
- (iii) Reducing the Null values.
- (iv) Disallowing the possibility of generating spurious tuples.

left and right side of FD is called determinants.

day / date:

Q. Define functional Dependency.

relation

→ it describes the relationship b/w attributes in a relation.

If A & B are attributes of relation R ,
 B is functionally dependant on A ,
if each value of A is associated
to exactly one value of B in R .

written as. ($A \rightarrow B$)

Main characteristics of functional dependency used in normalization.

(i) have 1:1 relationship b/w attributes on right & LHS.

(ii) holds $\forall$ time.

(iii) are non trivial.

Q. Inference RULES for Functional Dependency.

following are the six rules for functional dependencies.

(Armstrong's inference Rules).

* IR1 (reflexive) : If $Y \subseteq X$ then $X \rightarrow Y$
IR2 (Augmentation) : If $X \rightarrow Y$ then $XZ \rightarrow YZ$.
IR3 (Transitive) : If $X \rightarrow Y$ and $Y \rightarrow Z$
then $X \rightarrow Z$.
by completeness property.

IR4 (Decomposition) If $X \rightarrow YZ$ then
 $X \rightarrow Y$
 $X \rightarrow Z$.

IR5 (union) If $X \rightarrow Y$ and $X \rightarrow Z$

* Pseudo-transitive IR) If $X \rightarrow Y$ and $WY \rightarrow Z \Rightarrow XW \rightarrow Z$



Closure example:

I. $R(ABCDEFGG)$

- ① $A \rightarrow B$
- ② $BC \rightarrow DE$
- ③ $AE \rightarrow G$

$(AC)^+$

$$\begin{aligned}(AC)^+ &= AC \\ &= ABC \\ &= ABCDE \text{ Ans!}\end{aligned}$$

II. $R(ABCDEFGH)$

- ① $A \rightarrow BC$
- ② $CD \rightarrow E$
- ③ $E \rightarrow C$
- ④ $D \rightarrow AEH$
- ⑤ $ABH \rightarrow BD$
- ⑥ $DH \rightarrow BC$

then;
does $BCD \rightarrow H$ exist?

$$\begin{aligned}(BCD)^+ &= BCD \\ &= BCDE \\ &= ABCDEH\end{aligned}$$

FINDING CANDIDATE KEYS:-

hence yes!

I. $R(A, B, C, D, E, F)$

$$FD = \left\{ \begin{array}{l} AB \rightarrow C \\ C \rightarrow D \\ C \rightarrow E \\ E \rightarrow F \\ F \rightarrow A \end{array} \right\}$$

* Only those closure will be CK that return full table in Result.

— (a)

$$A^+ = A \text{ (@ ' . no)}$$

$$E^+ = AEF \text{ (no)}$$

$$B^+ = B \text{ (@ ' . no)}$$

$$F^+ = AF \text{ (no)}$$

$$C^+ = ACDE \text{ (@ ' . no)}$$

$$D^+ = D \text{ (" ")}$$

$$AB^+ \Rightarrow ABCDEFA \text{ (Yes!)}$$

hence a CK.

Similarly $A^+ \neq B^+$ are also candidate key
 $CK \{ AB, BC, AC, BF, BE \}$.

III. Solving through dependancy graph!

$R(A, B, C, D, E, F, H)$

$A \rightarrow D$

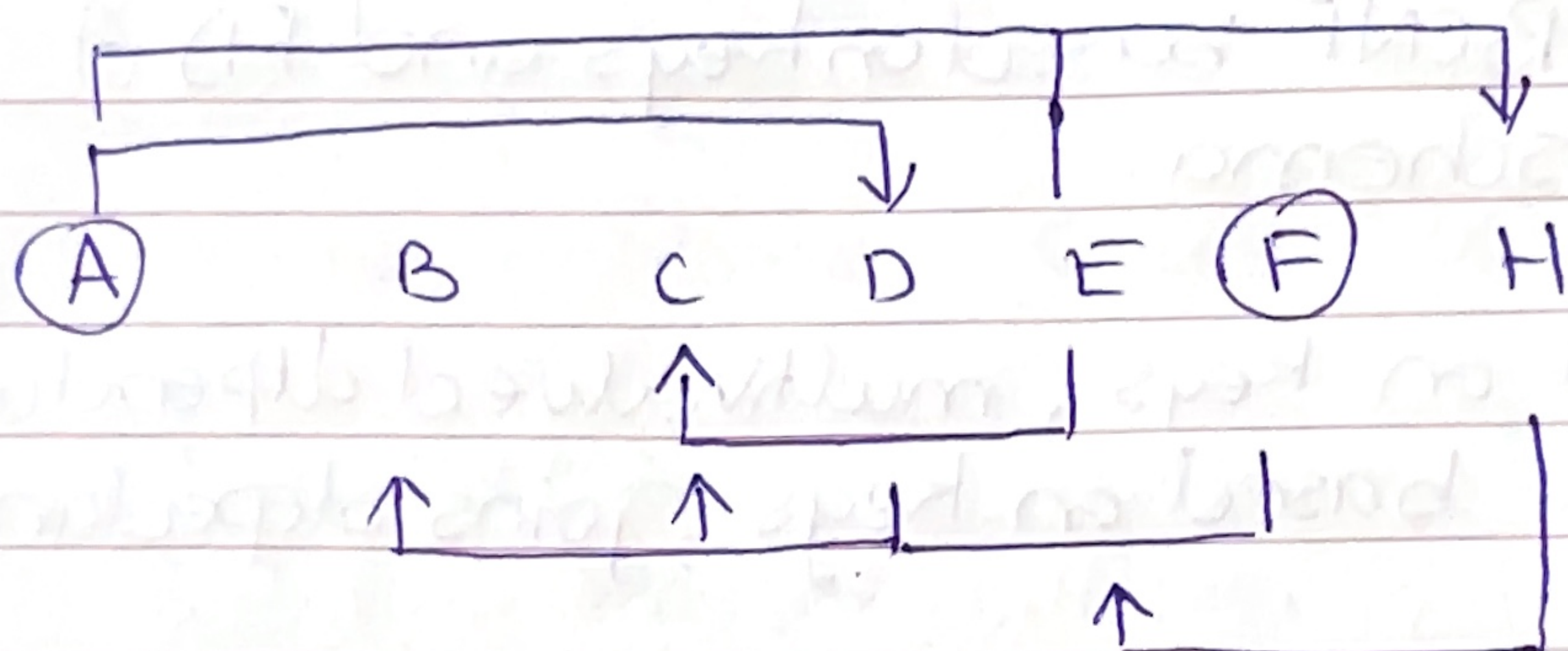
$AE \rightarrow H$

$DF \rightarrow BC$

$E \rightarrow C$

$H \rightarrow E$.

Now as per the graph
 (A) & (F) have no incoming
 arrow $\therefore$ they will
 always be the part of
 the CK.



BC cannot be part of the CK as they are not on the
 left hand side of FD .

Trying combinations of AF

$AF^+ : AFD$

$ABCDF \quad X$

$ABF^+ : X$

$ACF^+ : X$

$ADF^+ : X$

$AEF^+ : \checkmark$

$AHF^+ : \checkmark$

$\therefore CK \{ AEF, AHF \}$

NORMALIZATION:-

The process of breaking up or decomposing unsatisfactory / bad relations by breaking up their attributes into smaller parts.

Normal: Condition using keys and FDs of a relation to certify whether a relational schema is in a particular normal form

* 2NF, 3NF, BCNF based on keys and FD of a relational schema.

* 4NF Based on keys, multivalued dependencies:
* MVDs, SNFs based on keys, joins dependencies.

Additional properties may need to be ensuring of good relational design.

1ST NORMAL FORM:-

Rule:

The table should not contain any multivalued attribute.

↓
i.e each cell
should contain
single value.

Roll No	Name	Course
1	Sai	C/C++
2	Harsh	Java
3	Waqar	C/DBMS
4	Hari's	DSA.

> issue. ↓

each record
should be
unique.

Solution 1.

~x~

Roll No.	Name	Course
1	Sai	C
1	Sai	C++
2	Harsh	Java
3	Waqar	C
3	Waqar	DBMS
4	Hari's	DSA

primary key:

{ Rollno + Course }

↓

composite
primary key.

Solution no. 2

Roll No.	Name	Course 1	Course 2
1	Sai	C	C++
2	Harsh	Java	NULL
3	Waqar	C	DBMS
4	Hari's	DSA	NULL

primary key

{ rollno. }

full functional dependancy means removal of any attribute say A from X (if $X \rightarrow Y$) means that dependancy does not hold any more.

Solution no. 3

~X~

Roll No	Name
1	Sai
2	Harsh
3	Waqar
4	Haris

Roll No	Course
1	C
1	C++
2	Java
3	C
3	DBMS
4	DSA

Primary key
"Roll no."

Primary key =

"Roll no + Course"

2ND NORMAL FORM:-

2 Rules: ① Table or relation must be in 1NF.

② All non prime attributes of the table should be fully functionally dependant on the candidate key.

(i.e. no partial dependancy).

non prime attribute: attributes that are not the part of the candidate keys.

① Example ①:-

CustomerID	StoreID	Location
1	1	Delhi
1	3	Mumbai
2	1	Delhi
3	2	Bangalore
4	3	Mumbai

candidate keys:

Customer ID StoreID

Prime attributes $\Rightarrow$ Ck.

customer
storeID

non prime
attributes:
location.

transitive dependancy occurs when nonprime attribute depends upon a non-prime attribute rather than directly on the CK.

day / date: Only 1

But location in a table is dependant on storeID right? $\therefore$ the table is to be converted into 2nd normal form.

Solution

To convert it in 2NF (divide the table).

StoreID	Location

customerID	StoreID

dividing it in two tables we satisfied the fully functionally dependant property as the location in its own new table only depends on storeID which is the only candidate key of that table.

3rd NORMAL FORM:-

Rule:

The table should be in the 2nd Normal form.

There should be no transitive dependancy in the table. nonprime $\rightarrow$ nonprime attribute.

Roll no.	State	City.
1	Punjab	Mohali
2	Haryana	Ambala
3	Punjab	Mohali
4	Haryana	Ambala
5	Bihar	Patna.

Analysis
✓

CK = { Roll no. } / prime attribute.

Roll no $\rightarrow$ state

state $\rightarrow$ City. (Transitive dependancy).



partial dependency.
L.H.S subset of C.K. (AND) R.H.S is a nonprime attribute. day / date:

checking transitive dependency.

L.H.S must be a candidate key (or) R.H.S is a prime attribute.

BOYCE Codd NF (Special form of Normal form). ^{3rd}



More restriction!

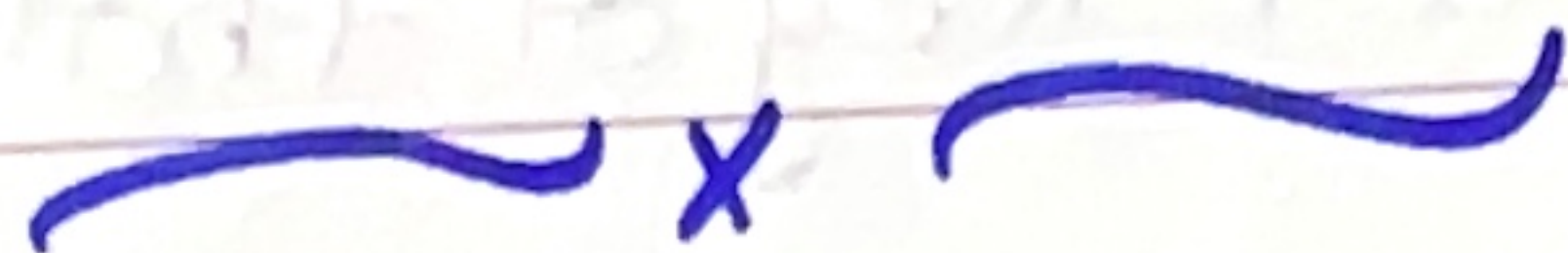
Rule:

(should be)

Every relation in 3NF

+ L.H.S should always be candidate key in the FD.

i.e no nontrivial functional dependency occurs.



TRANSACTIONS:-

A unit program Execution that access and (or) possibly updates various data items.

Two main issues to account for:

- (i) failure of various kind
- (ii) concurrent execution of various transactions.

∴ transactions should have atomicity property
i.e it will be completed altogether or never at all.

∴ once the transaction has been successfully completed with no failure at run time, we or transaction should ensure that update to the database must occur / persist even with any software / hardware failure. i.e durability property.